Thermal Simulation Report

Siliconware Precision Industries Co., Ltd

SPIL Internal Use

31x31 MCM(3)-PBGA 720

(Device: A9126LCB4A, 4L (1/2/1) Sub, 4L/6L PCB)

RTS14173, Aug. 19, 2014

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Customer	**Prepared by**	**Approved by**	**Code**	**Issue Date**
RDC	Freedman Yen	Nicholas Kao	Ansys	Aug 19, 2014

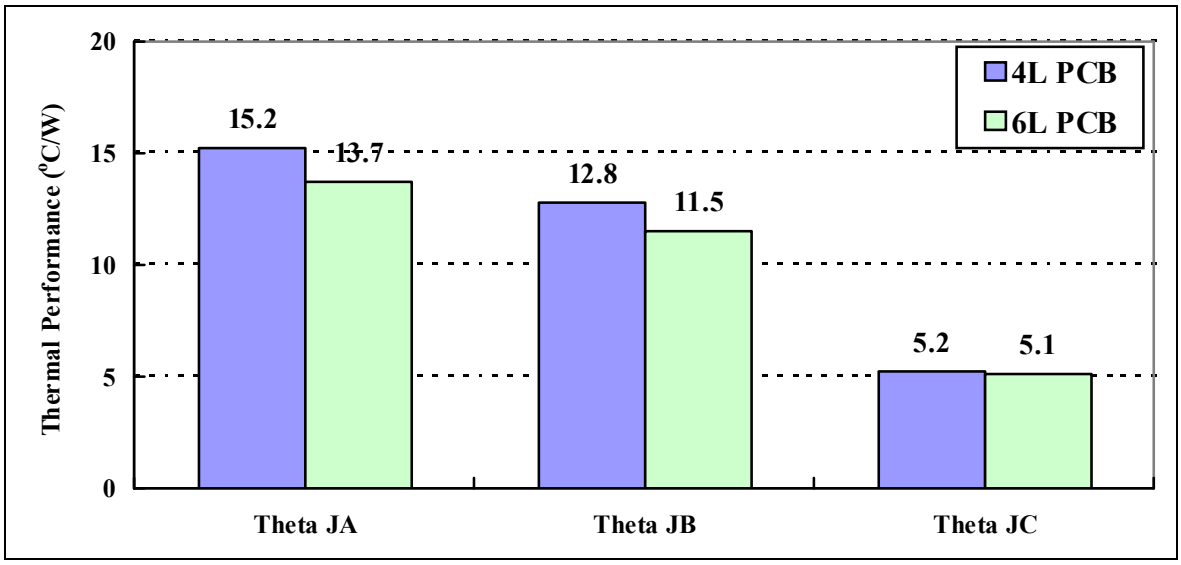
Siliconware Precision Industries Co., Ltd
NO. 123, Sec 3, Da Fong Rd, Tantzu, Taichung, Taiwan
Phone: +886-4-25341525
Fax: +886-4-25339099
http://www.spil.com.tw

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1. Introduction

In this report, we present the thermal characteristic for 720-pin MCM(3)-PBGA mounted on 4/6-layers PCB using the Finite Element Modeling (FEM) method. The junction-to-ambient thermal resistance is utilized to evaluate thermal performance for this package and then to predict the power dissipation capability by given ambient and maximum junction temperatures.

2. Result Summaries



Thermal Performance of MCM(3)-PBGA under Natural Convection

3. Conclusion

From the simulation results of temperature contours, we can obtain the chip junction temperatures (T_J) and the case temperatures (T_C). Since the input power (P) and ambient temperature (T_A) are controlled variables, we can calculate the shown above thermal resistance of θ_{JA} according to $\theta_{JA} = \frac{T_J - T_A}{P}$ and assuming ambient temperature of 25°C and maximum junction temperature of 125°C, we can predict the power dissipation capability of this package as below table.

PCB layers	4L	6L
Power Dissipation (watt)	6.58	7.3

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4. Assembly Descriptions

Package	Type		MCM(3)-PBGA 720
	Device Name		A9126LCB4A
	Substrate Drawing No.		I-W946 4L(1/2/1) Rev. A1
	Dimension (L x W)		31 x 31 mm
	Chip Size	U1	6.47 x 5.93 x 0.305 mm (254.7 x 233.5 x 12 mil)
		U2	6.67 x 6.17 x 0.305 mm (262.6 x 242.9 x 12 mil)
		U3	4.06 x 2.12 x 0.305 mm (160.1 x 83.5 x 12 mil)
	Ball Pitch		1.0 mm
	Number of Cu Layer-Substrate		4 layers
PCB	PCB Dimension (L x W)		4” x 4.5”
	PCB Thickness		1.6 mm
	Number of Cu Layer-PCB		4/6 layers
	PCB Number		N/A

5. Thermal Modeling Method

Thermal resistance represents the capability of an IC package to carry out the heat inside an IC chip. It is a complex function of package structure, material property, input power, environment variables such as air flow speed and PCB layers that can be analyzed by finite element modeling (FEM) techniques. In this simulation a commercial Finite Element Analysis (FEA) program, ANSYS, was used to create the 3D thermal model stacked by brick elements. Considering the package structure symmetry, only the quarter package structure was modeled to save CPU calculating time. All the materials are assumed isotropic with constant physical property and all material interfaces are also assumed perfect joined. The heat flux is applied on the top surface of the die to simulate heating power and constant convection coefficients applied on the exterior surface of this model. The physical properties and simulation conditions are summarized as below.

A. Simulation Conditions

Input Power	U1: 6 W , U2: 1.9 W U3: 0.1W
Test Board (PCB)	4 / 6 layers
Control Condition	Air Flow = 0 m/s

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B. Material Property

Item		Material	Thermal Conductivity K (W/m-k)
Package	Die	Si	147
	Substrate	Cu	400
		HL832NXA	0.8
	Silver Paste	AB2000	1.2
	Mold Compound	9750	0.9
	Solder Ball	Sn/Ag/Cu	55
PCB		Cu	400
		FR4	0.2

Appendix 1-Terminology

The major thermal dissipation paths can be illustrated as following

***T_J*: the maximum junction temperature;**

***T_A*: the ambient or environment temperature;**

***T_C*: the maximum compound surface temperature;**

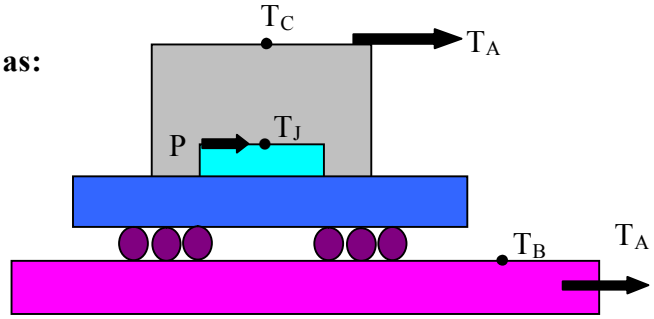
***T_B*: the surface temperature of PCB;**

***P*: total input power**

The thermal parameter can be define as following

1. *Junction to ambient thermal resistance, θ_{JA}*, **defined as:**

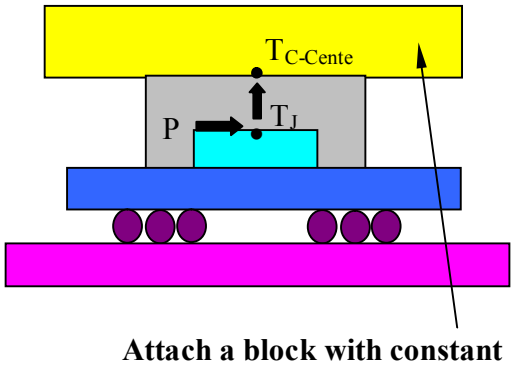
$$\theta_{JA} = \frac{T_J - T_A}{P} \ ; \qquad (1)$$



Thermal Dissipation of PBGA package

2. *Junction to case thermal resistance, θ_{JC}*, **defined as:**

$$\theta_{JC} = \frac{T_J - T_C}{P} \ ; \qquad (2)$$

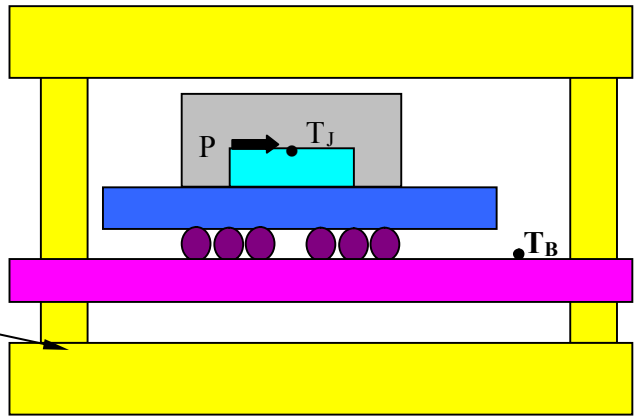


Attach a block with constant temperature onto package.

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3. *Junction to board thermal resistance,* θ_{JB}**, defined as:**

$$\theta_{JB} = \frac{T_J - T_B}{P} \; ; \qquad (3)$$



Attach a block with constant temperature

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